Roll Number: _____

Thapar Institute of Engineering and Technology, Patiala

Computer Science and Engineering Department

B E- CSBS (III Semester) EST	**Course Code: UCT303** **Course Name: Object Oriented Programming**
Dec 6, 2023, 2:00 PM	
Time: 3 Hours, M. Marks: 35	Name of Faculty: Dr. Nidhi Kalra

Note: Attempt any 5 questions out of 6 with proper justification. Assume missing data, if any, suitably.

Q1. **a)** Write a C++ code to create a template class called Bubble with following data member T a [25].
The Bubble class has the following public function declarations:

Function	**Description**
void get (int n)	This function gets the elements for the array(a). Print the sentence "Enter the array elements" in the function.
void display (int n)	This function displays the elements in the array.
void sort (int n)	This function sorts the array in ascending order.

Define the functions with the class template (template <class T>). In the main (), ask the user to enter the size of array and call all the methods by the Bubble class object of respective types (int/float). **(4 marks)**

b) Write a C++ code to create a vector (array) of 6 integer elements {10, 20, 10, 30,40,50}, and use the inbuilt function of algorithm class to count the occurrence of 10 in the array. **(3 marks)**

Q2. **a)** Amutha is a Kindergarten English teacher. She wanted to test her students after teaching the word game with comparison and concatenation. So, she has decided to conduct a viva voice to her students. She has given the option to the students; they can answer either concatenation or comparison. To conduct this test, she needs to write a C++ program using the binary operator overloading concept. To implement this: Create a class named MyString with data member char str[100] and the following public methods:

- void getData (): Reads an input string from the user.
- void display (): Displays the resultant string.
- int operator == (MyString str1): Compares two string using binary operator overloading.
- void operator+ (MyString str1): Concatenates two strings using binary operator overloading.

In main () use the switch case to take input for choosing concatenation or comparison and execute the above functions of the class. **(4 marks)**

[P.T.O]

b) Write a C++ code to convert one User defined data type -polar system to another User defined data type cartesian system. In main, create two classes (polar and cartesian), one object polar p(10,5); and execute the statements: cartesian c = p; c.show(); **(3 marks)**

Q3. **a)** Joey has to buy a gift for his friend's birthday. The minimum amount he can spend is Rs. 200 and maximum is Rs. 300. He decided to split the amount equally and buy two gifts for him instead. He then goes to the gift shop, picks up two items checks its cost. If the cost of both the items lies between 100-150 then only, he can pair the items as a gift. Create a class named Item with following private attributes: string name, int cost. Include the necessary getdata() and display() methods for the above mentioned class. Include a function: int operator && (Item &item) that will overload the && operator and return the int value to main () whether both the costs are between 100 and 150. **(4 marks)**

b) Write a C++ program to illustrate function template with non-type parameter? **(3 marks)**

Q4. **a)** Write a C++ program that reads an integer array from a text file named "input.txt" and calculates the sum of all elements. The program should then write the sum to another text file named "output.txt". **(4 marks)**
b) Explain the following terms in context to file handling in one line only. **(3 marks)**

i. ios::out
ii. ios::app
iii. ios::ate
iv. ios::binary
v. ios::trunc
vi. ios::cur

Q5. **a)** Write a C++ program that reads an array of integers from a binary file named "abc.dat". Sort the array in ascending order and write it back to the same binary file. **(4 marks)**
b) Write a C++ code to explain the functions seekg (), seekp (), tellg () and tellp (). **(3 marks)**

Q.6 **a)** Write a C++ programs to implement (i) virtual function (ii) pure virtual function. **(5 marks)**

b) Explain the difference between Function overloading and Function overriding with the help of suitable programs. **(2 marks)**